MIL-F-35054B

11 June 1976
SUPERSEDING
MIL-F-35054A
13 August 1965

MILITARY SPECIFICATION

FRUIT, CANDIED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers candied fruits for use in the galleys, kitchens and bakeries of the Armed Forces.

1.2 Classification. The candied fruit shall be of the following types and classes for type V, as specified (see 6.1):

TYPES

I - Cherries
II - Citron
III - Lemon peel
IV - Orange peel
V - Pineapple
Class 1 - Natural color
Class 2 - Green colored
VI - Mixed fruits

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-26 - Bag, Plastic, (General Purpose)
PPP-C-96 - Cans, Metal, 28 Gage and Lighter
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-G-460 - Glass Containers, One Gallon Capacity and Smaller, for Other Than Medicinal Products; Packaging and Packing Of

FSC 8925

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking For Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U. S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Applications for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. All material shall be of edible grade and shall be clean, sound and wholesome, and free from insect infestation or other objectionable foreign matter. The fruit shall be suitable in composition, appearance, size and texture for the applicable type.

3.1.1 Cherries. Cherries shall be of a suitable variety and shall be pitted.

3.1.2 Citron. Citron shall be true citron, the fruit of Citrus medica, and not of the melon variety.

3.1.3 Lemon peel. Lemon peel shall be the peel from sound ripe lemons, including the outer skin containing the natural yellow color and lemon oil.

3.1.4 Orange peel. Orange peel shall be the peel from sound ripe oranges, including the outer skin containing the natural orange color and oil of orange.

3.1.5 Pineapple. Pineapple shall be in whole slices with the core and rind removed.

3.2 Finished product. Each type of candied fruit shall possess a uniform characteristic color of the respective type. The surface of each type shall be smooth and not wrinkled. The finished product shall be free from any undesirable odor or flavor of any origin.

* 3.2.1 Type I. Cherries shall be thoroughly processed with sugar and corn sirup. They shall contain not more than 100 parts per million (ppm) of sulfurous acid calculated as sulfur dioxide, 75 ± 1 percent soluble solids and shall be artificially colored a bright red with a non-bleeding dye approved by the Federal Food and Drug Administration.

* 3.2.2 Type II. The citron shall be properly processed to yield the characteristic flavor, color and texture. The citron shall be thoroughly processed with sugar and corn sirup and shall contain 73 ± 1 percent soluble solids. It shall contain not more than 0.3 percent sodium chloride and not more than 100 parts ppm of sulfurous acid calculated as sulfur dioxide. Artificial coloring shall not be used.

* 3.2.3 Type III. The lemon peel shall be properly processed to yield the characteristic flavor, color and texture. The peel shall be thoroughly processed with sugar and corn sirup and shall contain 73 ± 1 percent soluble solids. The peel shall contain not more than 0.3 percent sodium chloride and not more than 100 ppm of sulfurous acid, calculated as sulfur dioxide. Artificial coloring shall not be used.

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* 3.2.4 Type IV. The orange peel shall be properly processed to yield the characteristic flavor, color and texture. The peel shall be thoroughly processed with sugar and corn sirup and shall contain 73 ± 1 percent soluble solids. The peel shall contain not more than 0.3 percent sodium chloride and not more than 100 ppm of sulfurous acid, calculated as sulfur dioxide. Artificial coloring shall not be used.

* 3.2.5 Type V, class 1. Class 1, pineapple, prepared in accordance with 3.1.5 shall be thoroughly processed with sugar and corn sirup. Artificial coloring shall not be used. Class 1 pineapple shall contain not more than 100 ppm of sulfurous acid, calculated as sulfur dioxide and 74 ± 1 percent soluble solids. Class 1 pineapple shall be used in type VI, mixed fruit.

* 3.2.5.1 Type V, class 2. Class 2, pineapple, prepared in accordance with 3.1.5 shall be artificially colored bright green with a non-bleeding dye approved by the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder, and shall be thoroughly processed with sugar and corn sirup. Class 2 pineapple shall contain not more than 100 ppm of sulfurous acid, calculated as sulfur dioxide and 74 ± 1 percent soluble solids.

* 3.2.6 Type VI. Mixed candied fruit shall consist of approximately 20 parts, by weight, of each fruit (type I through IV and type V, class 1) specified in 3.2.1 through 3.2.5. Prior to formation into the mixed candied fruit blend, types II through V fruits shall be chopped in a dicer set to yield 3/8 inch (9 mm) pieces and type I fruit shall be sliced in a slicer to yield 1/4 inch (6 mm) slices. The finished individual pieces shall not exceed 5/8 inch (15 mm) in any dimension. Not more than 25 percent, by weight, of any one type of fruit shall be less than 1/8 inch (3 mm) in their minimum dimension. After chopping or slicing as indicated, the fruit shall be lightly mixed until evenly distributed to form the mixed candied fruit.

3.3 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.4 Workmanship. The product shall be processed in establishments meeting the requirements of MIL-STD-668. The product shall be prepared and processed with minimum delay between the various stages of production, consistent with good commercial practice.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Preacceptance inspection (plant sanitation inspection). The product shall be rejected if produced in plants not meeting the requirements of 3.4.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.3.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of cherries and citron. Determination of compliance with requirements for identity (variety) as contained in 3.1.1 and 3.1.2 shall be accomplished by review of invoices, labels or other valid documents.

4.3.1.2 Testing of caps. Testing of caps shall be as shown in table I. Test requirement for tin plate shall be lot average requirement. Nonconformance to tin plate requirements shall indicate unacceptable caps and use of such shall be cause for rejection of the involved quantity of finished product.

TABLE I. Testing of caps

Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref & rqmt paragraph
1 cap	Caps	S-1	Tin plate	Nearest 0.01 lb/ base box	5.1.1 4.4.2

4.3.2 Process examination. Examination shall be performed during processing to determine compliance with requirements contained in 3.2 (as applicable) as concerns procedures and controls. Records shall be maintained of temperature control associated with specific procedures. In addition, examination shall be performed as necessary to determine conformance to requirements of 3.1. Nonconformance to condition requirements or to procedure and controls as determined by actual examination or as reflected by records shall be cause for rejection of the involved finished product.

* 4.3.3 Examination of end item. Examination shall be in accordance with tables III through V and 4.3.3.1 through 4.3.3.3. The sampling plans and acceptable quality levels (AQL's) expressed as defects per hundred units, shall be as shown in table II.

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TABLE II. Sampling plans and AQLs

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major A	Major B	Minor	Total
III	1 primary container	Primary containers	I	0.25	1.5	---	4.0
IV	Contents of 1 primary container	Primary containers	S-2	---	---	2.5	---
V	Contents of 1 primary container	Primary containers	S-3	---	2.5	---	6.5

TABLE III. Examination of primary containers

Category Major A	Major B	Minor	Defect	Defects applicable to glass w/cap	Defects applicable to plastic bags	Defects applicable to level C pkg
101			Broken or cracked.	X		X
102			Tear, hole or open seal.		X	X
	151		Chipped or thin spot.	X		X
		201	Not clean.	X	X	X
	152		Closure cap not kind specified.	X		
	153		Closure cap not tight. 1/	X		
		202	Cap cross threaded.	X		
	154		Enamel on cap (interior or exterior) when required, missing, incomplete, or easily abraded by the fingernail.	X		

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TABLE III. Examination of primary containers (cont'd)

Category			Defect	Defects applicable to glass w/cap	Defects applicable to plastic bags	Defects applicable to level C pkg
Major A	Major B	Minor				
	155		Pitted rust on cap. 2/	X		
	156		Sealing compound of cap missing (when required).	X		
	157		Nomenclature missing, incorrect or illegible.	X	X	X
		203	Other information missing, incorrect or illegible.	X	X	X

1/ For determining tightness of closure for screw type cap, only the following is applicable: Place the primary container upright on top of a filled and closed fiberboard case. Grasp the side of the cap firmly with fingertips and turn cap gently counter-clockwise with a minimum of downward pressure for one complete revolution. If the cap turns and the container remains stationary, a defect shall be scored as closure not tight.

2/ Rust that can be removed by wiping with a soft cloth will not be considered as pitted rust.

TABLE IV. Examination of net weight 1/

Category	Defect
Minor	
201	Net weight less than 15-1/4 ounces (432 g).
202	Net weight less than 7-1/2 ounces (212 g).

1/ Report results to nearest 1/8 ounce (gram). Lot shall be rejected if the sample data indicates a lot average net weight less than the specified net weight.

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TABLE V. Examination of finished product

Category		Defect
Major B	Minor	
151		Candied fruits do not possess a uniform characteristic color, flavor and texture of the respective type or class.
152		The surface of each type not smooth (wrinkled).
153		Rancid, musty or other undesirable odor or flavor.
	201	One or more candied pieces of type VI product exceed 5/8 inch (15 mm) in any dimension. 1/
	202	More than 25 percent, by weight, of any one type of fruit in type VI product less than 1/8 inch (3 mm) in their minimum dimension. 1/
	203	Mixed fruit pieces not evenly distributed.
	204	The weight of each fruit on type VI not within ± 25 percent of the nominal requirement.

1/ Examine 2 ounces from a sample unit.

4.3.3.1 Examination of shipping containers (containing glass bottles). Examination of the filled and sealed shipping containers shall be in accordance with the applicable examination criteria contained in PPP-G-460.

4.3.3.2 Examination of shipping containers (containing plastic bags). Examination of the filled and sealed shipping containers shall be in accordance with examination criteria contained in the appendix of PPP-B-636. In addition the following defects shall be included in the table of examination: Major: Markings missing, incorrect or illegible. Minor: Pad missing or not completely covering the stitches. Major: Separators missing or not full height. Minor: Arrangement not as specified. Minor: Separators not constructed from the same material as the shipping container.

4.3.3.3 Examination of shipping containers for level C package. Examination of the filled and sealed shipping container shall be in accordance with the criteria specified in 4.3.3.2 except that only the defect for marking shall apply.

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* 4.3.4 Sampling procedure and acceptance criteria for testing of end item. The finished product shall be tested for sulfur dioxide, sodium chloride and soluble solids in accordance with 4.4.1. The sample for testing shall be a 6-ounce (170 g) composite derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of primary containers. Results shall be reported to the same unit or decimal place specified for the pertinent requirements. Any test failures shall be cause for rejection of the lot.

4.4 Tests. Test procedures and controls which differ from those specified herein, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.4.1 Chemical analyses. Unless otherwise specified, chemical analyses, when required, shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists:

Test	Chapter	Method
Candied fruits:		
Soluble solids 1/	Fruit and Fruit Products	Total Solids-In-soluble matter present and Water-Insoluble Solids
Sulfur dioxide	Preservatives and Artificial Sweeteners	Free Sulfurous Acid
Sodium Chloride 2/	Plants	Chlorine-volumetric I

1/ Shall be determined as the difference between total solids and solids insoluble in water.

2/ To calculate as sodium chloride, multiply percent chlorine found by 1.649.

4.4.2 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PACKAGING

5.1 Preservation-packaging. The product shall be packaged level A, B, or C, as specified (see 6.1).

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* 5.1.1 Level A. Eight ounces (227 g) or 16 ounces (454 g) of the required type product shall be filled into a glass jar. The jar shall be closed with a knurled screw cap, or hermetic friction-type seal closure with pry-off feature, made from not less than commercial 0.25 pound per base box electrolytic tin plate, and coated outside with a corrosion-resistant lacquer or enamel. The cap shall be coated inside with an enamel suitable for the product and shall be lined with a liner pad having a vinyl facing or any other combination of materials suitable for the product. The screw cap shall be secured in position with a shrunk-on cellulose band or a pressure-sensitive adhesive, plastic film 3/4 inch wide tape applied around the circumference at the juncture of the cap and the jar. The tape shall overlap itself a minimum of one inch. Alternatively, a twist-off closure coated with an enamel suitable for the product may be used. The inside of the twist-off closure shall have a suitable sealing compound to effect a hermetic closure.

5.1.2 Level B. Sixteen ounces (454 g) of the required type of product shall be packaged in a plastic bag constructed and closed in accordance with style 2 of PPP-B-26. The bag material shall be not less than 0.0015 inch (.0004 mm) thick.

* 5.1.3 Level C. Eight ounces (227 g) or 16 ounces (454 g) of the product shall be packaged in a manner to protect it from damage and deterioration during shipment from the supply source to the first consuming activity in CONUS. The contractor may use his standard practice when it meets these requirements.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing, glass jars. The product in glass jars shall be packed in accordance with the level A requirements of PPP-G-460.

5.2.2 Level B packing.

5.2.2.1 Glass jars. The product in glass jars shall be packed in accordance with the level B requirements of PPP-G-460.

* 5.2.2.2 Plastic bags. Twenty-four pounds (10.9 kg) of one type, packaged in plastic bags, shall be packed into a snug-fitting fiberboard box constructed, closed and reinforced in accordance with style RSC, V3c, V3s, or V4s of PPP-B-636. If the bottom flaps are closed by stitching, the bottom of the box shall be covered with a paperboard pad to protect the bags. Bags shall be protected from stitches in the manufacturer's joint by applying a strip of pressure-sensitive tape over the stitches.

* 5.2.2.2.1 Non-metallic reinforcement of shipping container. When specified (see 6.1 and 6.2) the shipping container specified in 5.2.2.2 shall be reinforced with non-metallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that two bands may be used, one lengthwise and one girthwise.

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* 5.2.3 Level C packing. The product, packaged as specified in 5.1, shall be packed in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Labeling and marking.

5.3.1 Labeling. Commercial labeling complying with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder shall be acceptable.

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with PPP-G-460, as applicable and MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required and class required when type V is specified (see 1.2).
(c) Applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When packing specified in 5.2.2.2.1 is required.

6.2 Intended use of non-metallic reinforcement of shipping containers. Packing specified in 5.2.2.2.1 is intended for use for transfer at sea and specific overseas operations.

6.3 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.14D/DSAR 4145.7.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Preparing activity:

Army - GL

Project No. 8925-0050

Review activities:

Army - MD
Navy - MC, MS
DP-SS

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DOCUMENT IDENTIFIER AND TITLE
MIL-F-35054B - Fruit, Candied

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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SUBMITTED BY *(Printed or typed name and address - Optional)*

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